President's office
REGINAL ADMINISTRATION AND LOCAL GOVERNMENT
NAMANGA SECONDARY SCHOOL
FORM THREE OPENING SCHOOL TEST-APRIL 2021
BASIC MATHEMATICS

TIME: **1:30 hours**

INSTRUCTIONS:

I. ANSWER ALL QUESTIONS

1.Make C the subject of the formula.

$$V=\sqrt{\frac{RC}{R-C}}$$

2. Determine the value of x if

$$\left(\frac{1}{1024}\right)^{x+3}=32^{x+3}$$

3.The sum of two consecutive numbers is 37. Find the numbers.

4.The H.C.F. of two numbers is 11 and their L.C.M. is 7700. If one of the numbers is 275, find the other number.

5.If a positive number is decreased by 10, then it is equal to 75 times its reciprocal. Find that number.

6. A certain triangle has a perimeter of 3082m. The shortest side measures 79m less than the middle side, and the longest side measures 377m more than the middle side. Find the lengths of the three sides.

7. The ratio of boys to girls in a class is 3:5. If the total number of students is 32,

a) how many more girls are there than boys?
b) What is the percentage of girls in the class?

8. For the function $f(x)= (-2+x)^2$, Find

a) axis of symmetry
b) Maxmum or minimum value.
c) Turning point
d) domain and range

9. X varies as the cube of Y and inversely as square root of Z, X = 6 when Y = 3 and Z= 25. Find

A. i)An expression connecting X,Y,Z
ii)X when Y = 7 and Z = 9
iii)Y when X = 8 and Z = 16

B)If Y is increased by 20% and Z is decreased by 36%, find the percentage increase in X

10. The table below show the masses of 40 students to the kilograms

Mass	31-40	41-50	51-60	61-70	71-80
Frequencies	5	2	20	x	10

a) Find the value of x
b) state the median class and median frequency
c) Calculate mode